

CEO
M. Bilal Javaid

Valid from: 2nd June 2026

Valid To: 8th June 2026

Feasibility Report

AI-Powered Smart Hydration & Wellness Wristband



Client: Brad

About
M. Bilal Javaid

[Link to Intro](#)

- 8+ years of experience in industrial design and electronics engineering.
- Expertise in PCB design, 3D modeling, and Prototyping.
- Specialized in embedded systems (STM32, ESP32, Raspberry Pi)
- Proficient in CAD, Rendering and DFM
- Builds practical, manufacturable tech solutions by uniting design and engineering.

Table of Content

Project Scope

Chapter 01: Research and Development

- Approach
- Components In-House Inventory
- Milestones and Deliverable
- Timeline
- Block Diagram
- Outcomes

Chapter 02: PCB Design

- Approach
- Schematic Design
- PCB Design
- Milestones and Deliverables
- Timeline
- Outcomes

Chapter 03: Mechanical Design

- Approach
- Conceptual Design
- Realistic Renders
- Milestones and Deliverables
- Timeline

Chapter 04: App Development

- Approach
- Milestones and Deliverables
- Timeline

WBS: Work Breakdown Structure

Budgeting

Timeline and Budget Justification

Portfolio

Project Scope

The project focuses on the end-to-end development of a premium smart wearable prototype designed for nightlife wellness, passive hydration awareness, and biometric-driven behavioral insights. The scope includes the design and assembly of a compact Fitbit-style wristband prototype integrating heart rate, SpO2, and galvanic skin response (GSR) sensing with Bluetooth Low Energy communication and intelligent haptic feedback functionality. The project also includes development of production-structured embedded firmware for real-time biometric acquisition, signal filtering, BLE communication, battery monitoring, fallback reminder logic, and low-power optimization. In parallel, a modern dark-themed iOS application will be developed in SwiftUI featuring live biometric dashboards, hydration reminder management, session tracking, recovery scoring, long-term wellness insights, and BLE device management. A lightweight Apple Watch companion application will additionally provide mirrored session tracking, haptic notifications, and live wearable status synchronization. The overall objective is to deliver a fully functional, wearable-quality prototype system with clean architecture, scalable software foundations, stable wireless communication, and a refined user experience suitable for investor demonstrations, future iteration, and continued product development.

Key Requirements

- Development of a compact Fitbit-style smart wearable prototype
- Integration of heart rate, SpO2, and GSR/sweat sensing systems
- Real-time biometric monitoring and BLE data transmission
- Intelligent haptic hydration reminder functionality
- Production-structured embedded firmware architecture
- Low-power operation and battery optimization implementation
- SwiftUI-based premium iPhone application development
- Live biometric dashboard with session tracking features
- Apple Watch companion application with synced session support

- BLE pairing, streaming, reconnection, and command handling
- Recovery scoring and long-term wellness insight features
- USB-C rechargeable battery integration with protection circuitry
- Wearable-focused enclosure and ergonomic sensor placement
- Complete source code, documentation, and architecture delivery
- Functional prototype validation and demonstration video creation

At the completion of the project, a fully functional AI-powered smart hydration wristband will be delivered, including the wearable hardware prototype, embedded firmware, PCB and Schematics, iPhone app, Apple Watch companion app, and complete BLE communication framework. The final package will also include documented source code, BOM, setup instructions, architecture documentation, and a demonstration showcasing real-time biometric monitoring, app communication, and haptic reminder functionality.



Chapter : 01

Research and Development



1.1 Approach

The Research & Development (R&D) phase of the Smart Hydration Wristband project will focus on validating the technical feasibility, wearable usability, sensing reliability, and system architecture required to develop a stable functional prototype. This phase will establish the foundation for all future hardware, firmware, and mobile development activities by experimentally validating biometric sensing, Bluetooth communication reliability, haptic feedback performance, enclosure ergonomics, and power consumption behavior. The project will proceed through a structured engineering workflow beginning with system architecture definition, component evaluation, sensor characterization, and wearable feasibility analysis before moving into firmware prototyping, BLE communication validation, and mobile integration testing.

The first stage of R&D will involve defining the overall system architecture, including MCU selection, sensor communication protocols, battery management strategy, and BLE data flow structure. Special focus will be placed on evaluating low-power BLE-capable microcontrollers such as the ESP32 due to their integrated wireless functionality and development flexibility. Sensor interfacing requirements for heart rate, SpO2, and GSR acquisition will be analyzed to ensure stable data capture under real-world movement conditions commonly experienced in nightlife environments.

The biometric sensing subsystem will then be experimentally validated. The MAX30102 sensor or equivalent optical biosensor will undergo wrist-placement testing to determine optimal positioning, ambient light shielding requirements, skin contact pressure, and signal stability during motion. Simultaneously, the galvanic skin response (GSR) subsystem will be researched extensively because it represents the most experimental component of the product. Electrode material selection, placement geometry, skin contact consistency, analog signal conditioning, noise filtering, and ADC acquisition behavior will be evaluated to determine realistic sweat-response and hydration trend estimation capability. Multiple filtering approaches including moving average filtering and baseline drift compensation will be tested to improve signal quality.

The haptic feedback system will also undergo performance testing to ensure vibration alerts remain clearly perceptible in noisy nightlife environments while maintaining low power consumption and user comfort. Different vibration motor profiles, PWM drive strengths, and

enclosure mounting methods will be evaluated to achieve effective tactile feedback without excessive wrist discomfort.

Power system R&D will include battery capacity estimation, current consumption profiling, charging circuit validation, and runtime optimization analysis. The target of achieving 6+ hours of active session runtime will be validated by measuring current draw under continuous biometric monitoring and BLE transmission scenarios. Battery protection circuitry and USB-C charging integration feasibility will additionally be verified.

Following hardware validation, firmware architecture prototyping will begin. Initial firmware modules will be developed to validate sensor acquisition timing, BLE communication stability, battery reporting, haptic control logic, and fallback reminder behavior during BLE disconnections. Firmware modularity and scalability will be prioritized to ensure future production-level maintainability.

Parallel mobile-side R&D activities will validate BLE communication with iOS devices using SwiftUI test applications. BLE pairing, data streaming latency, packet integrity, auto-reconnection handling, and real-time biometric visualization will be experimentally verified before full application development begins.

Finally, enclosure and wearability experimentation will be conducted to ensure that the device maintains stable skin contact while remaining comfortable, lightweight, compact, and visually aligned with premium consumer wearables. Internal component placement, strap ergonomics, sensor exposure windows, and thermal behavior will all be evaluated before transitioning into the full prototype development stage.

Phase 1: System Architecture & Technical Feasibility Analysis

This phase establishes the engineering foundation of the project by defining the overall hardware and software architecture, communication flow, power requirements, and subsystem dependencies.

- Define complete wearable system architecture
- Select BLE-capable microcontroller platform
- Define sensor communication interfaces (I2C, ADC, GPIO)

- Establish BLE communication structure and packet flow
- Determine power distribution architecture
- Evaluate firmware scalability requirements
- Define mobile app integration requirements
- Analyze wearable size and mechanical constraints
- Establish development and testing methodology

Phase 2: Biometric Sensor Research & Validation

This phase focuses on experimentally validating the heart rate, SpO2, and GSR sensing systems to determine signal reliability, placement requirements, and real-world usability.

- Integrate MAX30102 sensor for HR and SpO2 testing
- Evaluate wrist placement accuracy and stability
- Test motion interference impact on readings
- Research GSR sensing methodologies
- Test electrode placement configurations
- Evaluate skin contact consistency
- Develop analog signal conditioning approach
- Implement preliminary signal filtering techniques
- Validate ADC acquisition stability

Phase 3: Haptic Feedback & User Interaction Testing

This phase validates the effectiveness of the wearable's vibration feedback system in nightlife environments.

- Select low-profile vibration motor
- Test vibration intensity levels

- Evaluate haptic perception during movement
- Analyze comfort and usability impact
- Implement adjustable vibration patterns
- Optimize motor drive circuitry
- Validate response timing performance

Phase 4: Power System & Runtime Optimization

This phase focuses on validating battery life, charging reliability, and low-power operation strategies.

- Select rechargeable LiPo battery
- Measure active current consumption
- Evaluate BLE transmission power impact
- Test continuous sensor monitoring runtime
- Validate charging circuitry operation
- Implement battery percentage monitoring
- Analyze low-power firmware strategies
- Estimate total operational runtime

Phase 5: Firmware Prototype Development & BLE Validation

This phase establishes the embedded firmware framework and validates wireless communication stability.

- Develop sensor acquisition firmware
- Implement BLE communication services
- Test real-time data transmission
- Validate pairing and reconnection handling

- Implement haptic trigger commands
- Develop battery reporting logic
- Test fallback reminder behavior
- Validate firmware stability under continuous operation

Phase 6: Mobile App Communication & Wearability Validation

This phase validates iOS communication, wearable usability, and system integration.

- Develop preliminary SwiftUI BLE interface
- Validate live biometric data streaming
- Test session synchronization behavior
- Evaluate BLE latency and stability
- Conduct enclosure fit and comfort testing
- Validate sensor skin contact under movement
- Analyze wearable ergonomics and usability
- Verify complete subsystem integration

Component Inventory & Cost Table

Component	Quantity	Estimated Cost (USD)
ESP32 BLE MCU Module	1	\$12
MAX30102 HR + SpO2 Sensor	1	\$8
GSR Sensor Circuit Components	1 Set	\$15
Stainless Steel Electrodes	2	\$6
Low-Profile Vibration Motor	1	\$5
500mAh LiPo Battery	1	\$10
USB-C Charging Module	1	\$6
Battery Protection Circuit	1	\$4
Voltage Regulation Components	1 Set	\$5
Wiring & Connectors	1 Set	\$5
Wrist Strap Assembly	1	\$10
Miscellaneous Passive Components	1 Set	\$8
Testing & Debug Accessories	1 Set	\$10

Estimated Total Component Cost: \$104 USD

NOTE:

The Components listed in the component inventory are indicative and provided for preliminary planning and architectural understanding only. Final selection will be carried out during the detailed R&D phase based on validated functional requirements, power consumption analysis, form-factor constraints, availability, and system-level performance considerations.

DEVELOPMENT COSTING TABLE

Item	Estimated Cost
PCB Fabrication (1- Basic)	\$200
PCB Fabrication (3- Plus)	\$600
PCB Fabrication (5 -Premium)	\$1000
Mechanical Enclosure	\$100

Milestones and Deliverables

Milestones

- System architecture definition and component selection completed
- Biometric sensing subsystem experimentally validated
- GSR signal acquisition and filtering approach finalized
- Haptic feedback system validated for wearable use
- Battery runtime and charging system verified
- BLE communication and firmware prototype operational
- Preliminary iOS BLE communication successfully tested
- Wearability and enclosure feasibility validated
- Integrated subsystem functionality demonstration completed

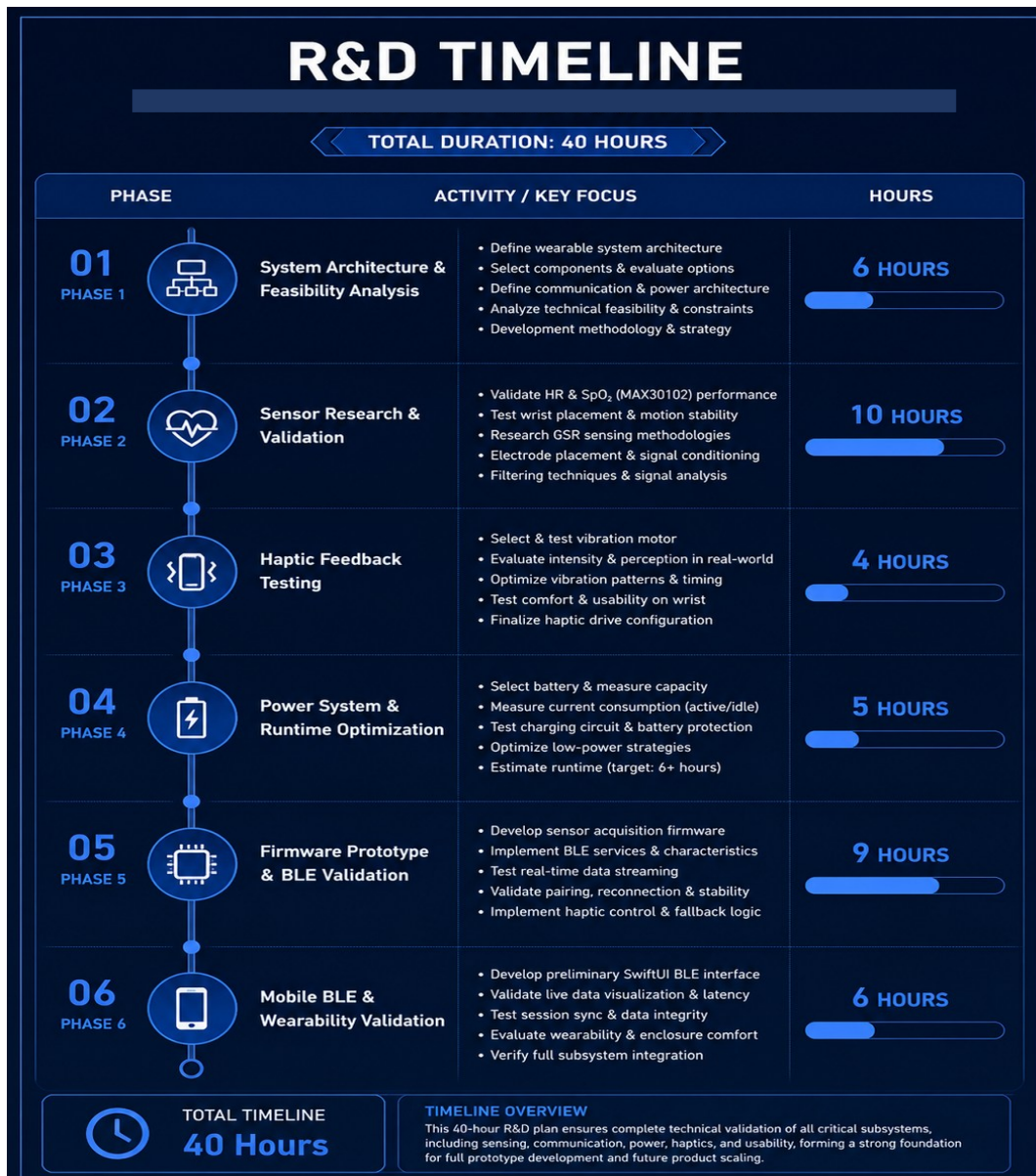
Deliverables

- Technical feasibility analysis documentation
- Wearable system architecture documentation
- Sensor validation and signal analysis reports
- Firmware prototype source code
- BLE communication framework validation
- Preliminary SwiftUI BLE test application
- Battery and runtime optimization report
- Wearability and enclosure assessment report
- Component BOM and integration documentation
- Functional subsystem demonstration setup

Outcomes

- Validation of wearable biometric sensing feasibility
- Stable BLE communication framework establishment
- Reliable haptic reminder implementation strategy
- Power-efficient wearable operation methodology
- Wearable comfort and sensor placement optimization
- Foundational firmware and mobile architecture readiness
- Reduced technical uncertainty before full prototype development

Timeline

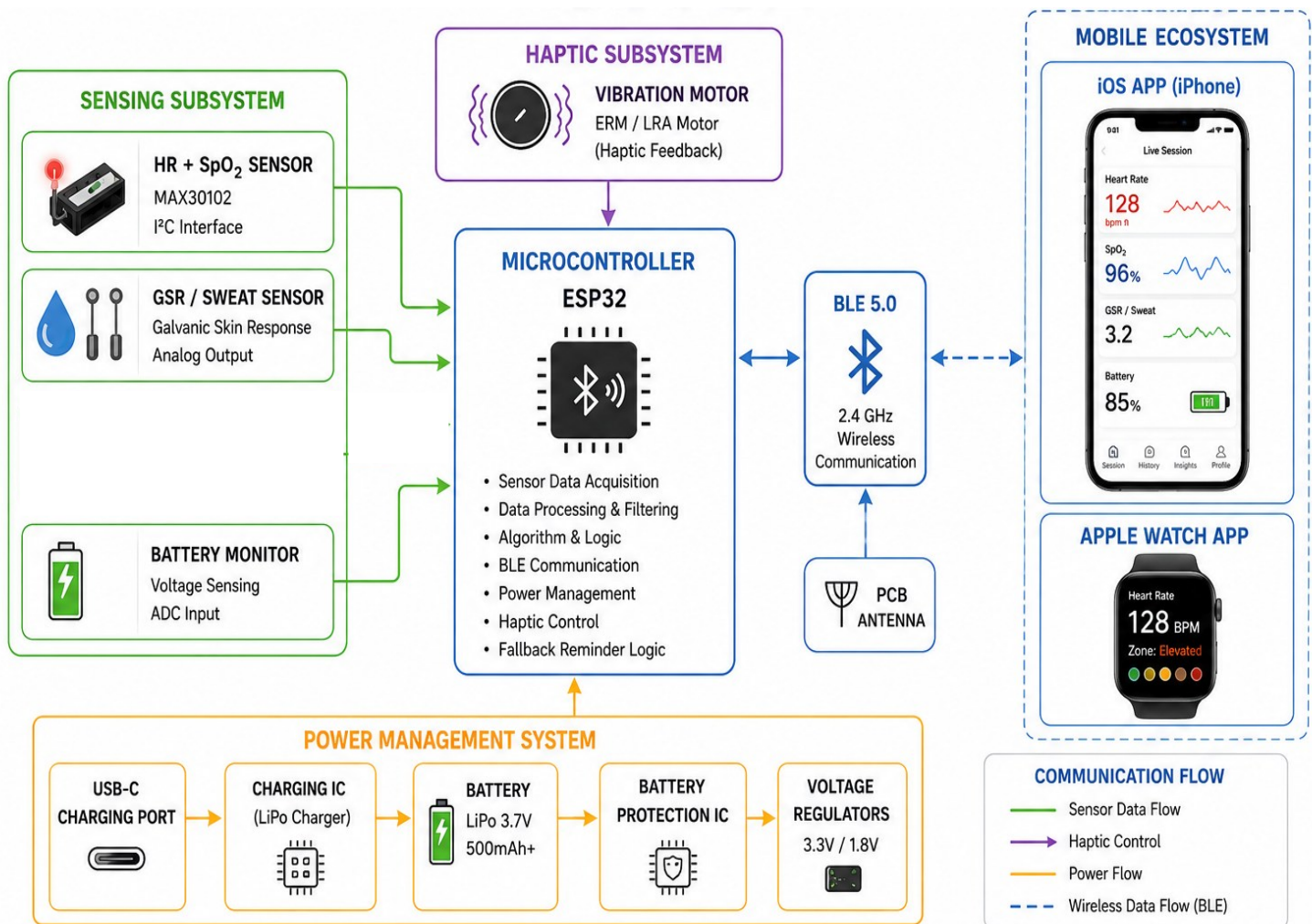


Total timeline for R&D: 40 Hours

Note

It is important to note that the component procurement, sourcing, and logistics processes are not included within the defined 40-hour R&D timeline. The provided schedule strictly covers engineering activities related to system analysis, debugging, optimization, and validation. Any delays or dependencies arising from component availability, supplier lead times, or procurement procedures fall outside the scope of this timeline.

Block Diagram





Chapter : 02



PCB Design



2.1 Approach

The PCB section of the Smart Hydration Wristband project focuses on transforming the validated R&D architecture into a compact, reliable, wearable-grade electronic hardware platform capable of supporting continuous biometric sensing, BLE communication, power management, and haptic feedback within a constrained wearable form factor. This phase will involve complete schematic development, component integration, PCB layout engineering, signal integrity optimization, power routing validation, manufacturability analysis, and prototype-ready design preparation. The PCB development process will proceed systematically from circuit-level architecture definition to final layout verification while ensuring the board remains compact, electrically stable, low power, and mechanically compatible with the wearable enclosure.

The PCB design workflow will begin with system-level hardware mapping based on the validated R&D findings. All critical subsystems including the ESP32 BLE microcontroller, MAX30102 biometric sensor, GSR sensing circuitry, battery charging system, USB-C power interface, vibration motor driver circuitry, voltage regulation network, and battery protection circuitry will be electrically integrated into a unified wearable hardware architecture. During this stage, component compatibility, voltage requirements, signal routing constraints, and communication interfaces such as I2C, ADC, GPIO, PWM, and BLE-related hardware considerations will be carefully analyzed.

Special attention will be given to the analog front-end design of the GSR subsystem because galvanic skin response acquisition is highly sensitive to noise, interference, and motion artifacts. Proper analog signal conditioning methods including filtering, impedance stabilization, grounding strategy, and ADC protection will be implemented to improve signal quality and maintain stable acquisition during wearable operation. Similarly, the MAX30102 optical sensor circuitry will require careful layout planning to reduce electrical noise and maintain stable heart rate and SpO2 sensing performance.

The power subsystem will be engineered to support efficient low-power operation while maintaining stable voltage regulation for continuous sensing and BLE transmission. Battery charging circuitry, LiPo protection implementation, USB-C power input routing, voltage regulators, current handling capability, and battery monitoring circuitry will all be validated to ensure safe and reliable operation. Thermal considerations and current consumption

optimization will also be evaluated to maximize runtime within the compact wristband form factor.

After schematic completion, the PCB layout stage will focus on compact component placement, wearable mechanical compatibility, signal integrity, and EMI reduction. Sensitive analog circuitry such as GSR and biometric sensing traces will be isolated from high-noise digital and power sections wherever possible. Ground plane design, decoupling capacitor placement, trace width optimization, impedance management, and vibration motor noise suppression techniques will be incorporated to improve overall system reliability.

Mechanical constraints of the wristband enclosure will additionally guide PCB dimensions, connector orientation, sensor exposure regions, mounting points, and battery positioning. The board layout will prioritize modularity and future iteration capability while maintaining a consumer-wearable appearance instead of a development-board-style implementation.

Finally, complete design rule checks (DRC), electrical rule checks (ERC), manufacturability verification, and prototype fabrication preparation will be performed. Final outputs will include schematic files, PCB layout files, BOM documentation, Gerber manufacturing files, assembly references, and hardware integration documentation required for prototype assembly and testing.

Schematics Section

The schematic development section defines the complete electrical architecture of the wearable system and establishes how all electronic components interact with each other. This stage is critical because it determines system stability, signal reliability, power integrity, sensor performance, and future scalability of the hardware platform. The schematics will act as the foundation for PCB layout development and hardware integration.

The schematic design process will involve detailed subsystem-level circuit development for biometric sensing, BLE communication, power regulation, battery charging, vibration control, and wearable interfacing. Each circuit will be validated for electrical compatibility, low-power operation, and manufacturability before transitioning into PCB layout design.

Phase 1: System Circuit Architecture Development

This phase establishes the complete circuit-level architecture of the wearable platform.

The objective of this phase is to define how all subsystems electrically interact while ensuring voltage compatibility, communication reliability, and low-power operation across the wearable platform.

- Define complete circuit architecture
- Map subsystem interconnections
- Define I2C, ADC, PWM, and GPIO interfaces
- Establish voltage domains and power flow
- Validate MCU peripheral allocation
- Plan sensor interface circuitry
- Define grounding and protection strategy

Phase 2: Biometric Sensor Schematic Design

This phase focuses on integrating HR, SpO2, and GSR sensing circuitry into the schematic architecture.

- Special emphasis will be placed on low-noise analog signal acquisition and stable sensor interfacing to ensure accurate wearable biometric monitoring.
- Integrate MAX30102 sensor circuitry
- Design GSR analog acquisition stage
- Add filtering and conditioning components
- Implement sensor protection circuitry
- Optimize ADC signal path
- Define sensor connector interfaces
- Validate signal integrity requirements

Phase 3: Power Management & Charging Circuit Design

This phase develops the wearable power subsystem and charging architecture.

- The goal is to ensure safe battery charging, efficient voltage regulation, and stable runtime performance while minimizing power consumption.
- Integrate LiPo charging IC
- Implement battery protection circuitry
- Design voltage regulation stages
- Add battery monitoring circuitry
- Integrate USB-C power input
- Validate current handling capability
- Implement decoupling strategy

Phase 4: BLE & Haptic Driver Circuit Integration

This phase integrates wireless communication support circuitry and vibration control electronics.

- Reliable BLE operation and stable haptic feedback control are essential for maintaining seamless user interaction and wearable responsiveness.
- Integrate ESP32 BLE circuitry
- Add antenna clearance planning
- Design vibration motor driver stage
- Implement PWM control circuitry
- Validate communication routing requirements
- Add debugging and programming interfaces

PCB Design

The PCB design section converts the validated schematics into a manufacturable wearable hardware layout optimized for compactness, signal integrity, low-power performance, and mechanical integration. This stage focuses heavily on routing optimization, analog isolation, wearable ergonomics, EMI reduction, and prototype manufacturability.

PCB layout development will prioritize wearable form factor constraints while ensuring stable operation of sensitive biometric sensing circuitry and BLE communication systems. Careful placement and routing strategies will be implemented to maximize reliability and future scalability.

Phase 1: Component Placement & Mechanical Planning

This phase establishes the physical organization of the PCB within the wearable

- Proper component placement is essential for wearable comfort, sensor contact stability, battery balancing, and signal integrity optimization.
- Define PCB dimensions and shape
- Position biometric sensors for skin contact
- Allocate battery and charging areas
- Optimize ESP32 placement for BLE performance
- Define connector orientations
- Plan mounting hole positions
- Align layout with enclosure structure

Phase 2: PCB Routing & Signal Integrity Optimization

This phase develops the electrical routing structure of the wearable PCB.

- Routing optimization is critical for minimizing electrical noise, maintaining analog signal stability, and ensuring reliable communication performance.
- Route power distribution network

- Route I2C and ADC signal traces
- Separate analog and digital sections
- Implement ground plane strategy
- Optimize trace widths and spacing
- Add decoupling capacitor placement
- Reduce EMI and switching noise

Phase 3: Design Verification & Manufacturing Preparation

This phase validates the PCB design and prepares it for fabrication.

- Verification ensures the PCB is electrically correct, manufacturable, and ready for assembly without major design risks.
- Perform ERC and DRC verification
- Validate footprint compatibility
- Generate Gerber manufacturing files
- Create assembly documentation
- Finalize BOM documentation
- Prepare fabrication package
- Review manufacturability constraints

Milestones and Deliverables

Milestones

- System-level schematic architecture finalized
- Sensor integration circuitry validated
- GSR analog acquisition stage completed
- Power management circuitry verified
- BLE and haptic driver integration completed
- Wearable PCB layout finalized
- Signal integrity optimization completed
- Design rule verification passed
- Manufacturing files generated successfully
- Prototype-ready PCB package completed

Deliverables

- Complete schematic design files
- Wearable PCB layout files
- Gerber manufacturing package
- Bill of Materials (BOM)
- Component placement documentation
- Assembly reference documentation
- Power architecture documentation
- Sensor integration documentation
- PCB design review report

- Manufacturing preparation package

Outcomes

- Compact wearable-grade PCB architecture
- Stable biometric sensing hardware platform
- Optimized BLE communication layout
- Low-noise GSR acquisition circuitry
- Reliable power distribution and charging system
- Manufacturable wearable PCB design
- Prototype-ready hardware documentation
- Improved hardware scalability for future revisions

Timeline



Note:

We can also provide PCB fabrication carried out through JLCPCB, ensuring high-quality manufacturing and adherence to design specifications. The estimated lead time for fabrication and delivery is approximately 2-3 weeks.



Chapter : 03



Mechanical Design

3.1 Approach

The mechanical design will start with an initial sketch phase, where the overall wristband shape, enclosure size, strap direction, underside sensor-contact area, USB-C port position, and internal component placement will be roughly defined. This phase will help establish the Fitbit-sized form factor and confirm how the device will sit on the wrist.

After the initial sketch is finalized, the design will move into the initial CAD phase, where the wristband body, top enclosure, bottom enclosure, strap attachment points, PCB space, battery pocket, haptic motor location, and sensor openings will be modeled in 3D.

Once the initial CAD is reviewed, the design will be refined into a final CAD model with proper wall thickness, rounded edges, screw bosses or snap-fit features, USB-C cutout, sensor alignment features, GSR electrode placement, and internal mounting supports. This final CAD will be prepared for 3D printing and assembly.

After final CAD approval, a physical prototype will be fabricated using 3D printing. The prototype will be used to test wrist comfort, sensor skin contact, strap fitting, USB-C access, battery placement, haptic vibration feels, and overall wearable usability. Any issues found during prototype testing can then be corrected in the next CAD revision.

• Important Design Aspects

The wristband must be compact, lightweight, comfortable, and suitable for real-world nightlife use. The enclosure should have rounded edges and a premium wearable appearance. The design should not look like a development board strapped to the wrist.

The bottom side is the most critical mechanical area because the heart-rate/SpO₂ sensor and GSR electrodes must maintain stable skin contact during movement. The GSR sensing area should be slightly raised or positioned so the electrodes touch the wrist properly without causing discomfort. The client specifically requires stable wrist readings, optimized sensor placement, and stable GSR skin contact.

The top enclosure will not include a display in this phase. This keeps the prototype thinner, lowers power consumption, reduces mechanical complexity, and makes the first wearable prototype easier to fabricate and test.

• Components Layout

The internal layout will include the following major components:

- BLE MCU board, such as ESP32 or equivalent
- HR/SpO₂ optical sensor module

- GSR/sweat-sensing electrodes on the underside
- GSR signal-conditioning circuit
- 500 mAh or higher LiPo battery
- USB-C charging module
- Battery protection circuit
- Haptic vibration motor
- Battery monitoring circuit
- Internal wiring or compact connectors

The PCB and electronics will be placed inside the central enclosure body. The battery should be placed flat inside the enclosure to keep the height low. The haptic motor should be mounted close to the wrist-contact side or mechanically coupled to the enclosure so the vibration can be felt clearly. The USB-C port will be positioned on the side wall of the enclosure for charging access.

• **Material Selection & Fabrication Strategy**

For the first prototype, the enclosure can be **3D printed**, as the client accepts a 3D-printed enclosure for Version 1. The aim is to validate fit, comfort, sensor contact, internal packaging, and wearable usability before moving toward custom PCB or molded enclosure development.

Recommended materials:

Part	Recommended Material
Top enclosure	SLA resin, PLA+, or PETG
Bottom enclosure	SLA resin, PLA+, or PETG
Strap	Silicone, TPU, or nylon
GSR electrodes	Stainless steel, copper with plating, or gold-plated pads
Fasteners	screws
Prototype finish	Sanding, primer, matte coating, or textured black finish

• **Mechanical & Component Mounts**

The enclosure will be split into a top shell and bottom shell.

The top shell will act as the outer cosmetic cover. It will be smooth, curved, and display-free. It can include a small logo area or tiny status LED window if needed, but no large screen cutout is required.

The bottom shell will include the sensor-contact region. It should have:

- Optical sensor window for HR/SpO₂
- Two GSR electrode contact pads
- Slight wrist-facing curvature
- Smooth edges for comfort
- Correct sensor pressure against the skin

Internally, the enclosure should include mounting bosses, screw posts, battery pocket, sensor alignment features, wire channels, and motor seating. Screws are preferred for the prototype because the enclosure may need to be opened repeatedly during testing.

• **Positioning & Integration**

The sensor module should be placed at the center of the wrist-facing bottom surface. This gives better contact and reduces movement-related signal noise.

The GSR electrodes should be positioned near the optical sensor area but electrically and mechanically separated. The electrodes should be large enough to maintain stable skin contact but not so large that they create discomfort.

The battery should be positioned flat inside the body to reduce thickness. The USB-C port should be located on the side, away from the wrist-contact surface. Strap attachment points should be designed at both ends of the enclosure and should support an off-the-shelf strap or custom silicone/TPU strap.

• **Prototyping & Testing**

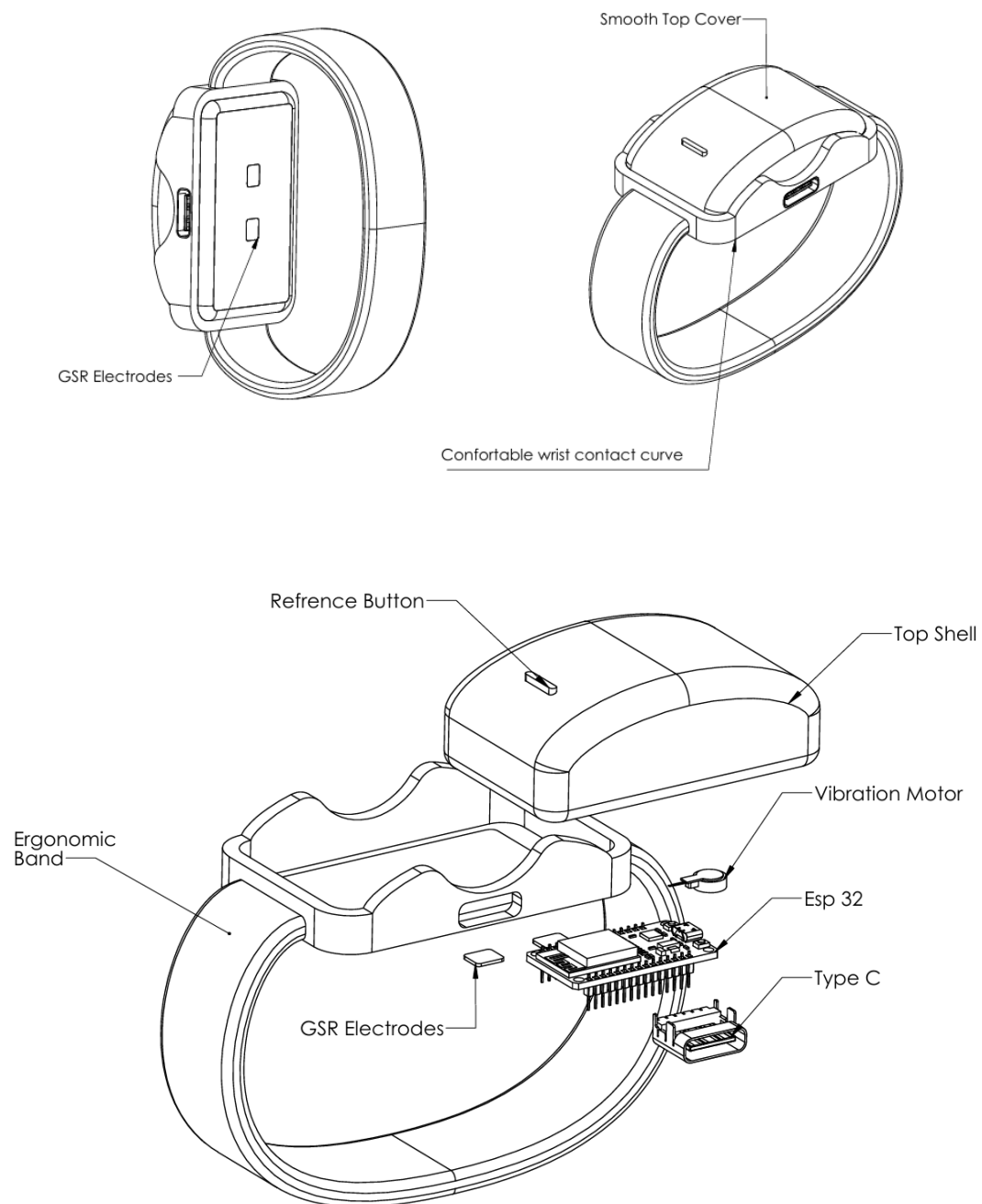
The first prototype should be tested for:

- Wrist comfort
- Strap tightness
- Sensor skin contact
- GSR electrode stability
- HR/SpO₂ sensor alignment
- Haptic vibration strength
- USB-C charging access

- Battery fit and safety clearance
- Internal component movement
- 6+ hour wearable usability

The main mechanical risk is poor sensor contact during wrist movement. Multiple bottom-shell iterations may be needed to tune the height of the optical sensor window and GSR electrodes.

Conceptual Design



Realistic Renders



Milestones and Deliverables:

Milestone 1: Mechanical Concept & Layout

Deliverables:

- Wristband body concept
- Internal component placement plan
- Sensor-contact underside concept
- Strap attachment direction

Milestone 2: CAD Design

Deliverables:

- 3D CAD model of top shell
- 3D CAD model of bottom shell
- Internal mounting features
- USB-C port cutout
- Strap connection geometry

Milestone 3: Prototype File Preparation

Deliverables:

- STEP files
- STL files for 3D printing
- Exploded view
- Assembly view
- Basic mechanical BOM

Milestone 4: Prototype Testing & Refinement

Deliverables:

- Fit and comfort review
- Sensor-contact validation
- Haptic motor placement review
- Updated CAD files after changes

Timeline



Total timeline for Mechanical Team: 60 Hours



Chapter : 04

App Development



4.1 Approach

1. Approach

The mobile application will act as the main user interface for the smart hydration wristband. The app will receive live biometric data from the wearable through Bluetooth Low Energy, display heart rate, SpO2, GSR/sweat trends, battery level, connection status, and hydration reminder activity.

For the MVP phase, the app should focus on validating the core product experience: starting and ending hydration sessions, showing live biometric readings, managing BLE connection with the wristband, sending haptic reminder commands, storing session history, and supporting a basic Apple Watch companion experience.

The app can be developed using SwiftUI for iPhone and WatchOS to match the client's requirement for a modern iOS-native experience. The design should feel premium, dark-themed, lifestyle-focused, and closer to Oura/Whoop-style wellness apps rather than a medical dashboard.

Development Phases

Phase 1: Requirement Finalization & App Architecture

This phase will define the app scope according to the wearable device functionality and MVP goals.

Main focus:

- Define app flow for onboarding, pairing, dashboard, session tracking, and settings.
- Define BLE communication flow between wristband and iPhone app.
- Define active session start/end behavior.
- Define live biometric data display.
- Define haptic reminder logic from app to wearable.
- Define battery and connection status workflow.
- Define Apple Watch companion app scope.
- Define local data storage for sessions and recovery ratings.

Phase 2: UI/UX Design

The app interface should look premium, modern, and simple. Since the product targets nightlife wellness, the UI should use a dark, clean, and lifestyle-focused design instead of a medical-style layout.

Main focus:

- Onboarding screens.
- Device pairing screen.
- Main dashboard.
- Start/end session screen.
- Live biometric dashboard.
- Heart rate zone visualization.
- GSR/sweat trend display.
- Battery and connection status.
- Hydration reminder activity screen.
- Session history screen.
- Morning recovery rating screen.
- Basic settings screen.
- Apple Watch companion screen layout.

Phase 3: iOS SwiftUI App Foundation

This phase will create the base iPhone app structure and prepare it for BLE, session tracking, and biometric data visualization.

Main focus:

- SwiftUI project setup.
- Modern iOS architecture.
- Navigation structure.
- Reusable UI components.
- Local storage setup.
- Session data model.
- User settings model.
- Mock biometric data mode for early testing.
- Permission handling for Bluetooth and notifications.
- Basic chart/data visualization components.

Phase 4: BLE Communication Layer

The app will communicate with the wristband using Bluetooth Low Energy. The wearable will stream biometric data to the app, and the app will send commands such as haptic reminder triggers and session control.

Main focus:

- BLE device discovery.
- Pairing and connection handling.
- Auto-reconnection handling.
- Receive heart rate data.
- Receive SpO2 data.
- Receive GSR/sweat data.
- Receive battery percentage.
- Receive device status.
- Send haptic reminder command.
- Send session start/end command if required.
- Prepare BLE UUID and characteristic documentation.

Example BLE data packet:

```
{  
  "device_id": "HYDRO-WR-001",  
  "heart_rate": 92,  
  "spo2": 97,  
  "gsr": 438,  
  "battery": 78,  
  "session_active": true  
}
```

Example app-to-wristband command:

```
{  
  "command": "trigger_haptic",  
  "pattern": "hydration_reminder",  
  "duration": 3  
}
```

Phase 5: Session Tracking, Recovery Rating & Insights

This phase will implement the main wellness experience of the app. The user can start a nightlife session, view live biometric trends, receive hydration reminders, and later rate their morning recovery.

Main focus:

- Start session.
- End session.
- Store session duration.
- Store biometric averages.
- Store hydration reminder count.
- Display live HR, SpO2, and GSR values.
- Display HR zone status.
- Save session history locally.
- Morning recovery rating from 1–10.
- Correlate recovery rating with previous session data.
- Basic long-term insight screen.
- Basic trend summary for user behavior.

Phase 6: Apple Watch Companion App

The Apple Watch app will provide a lightweight companion experience. It does not need to duplicate the full iPhone app but should support quick visibility and haptic feedback.

Main focus:

- Mirror active iPhone session.
- Show basic session status.
- Show live HR zone status.
- Show connection/session state.
- Deliver haptic hydration reminders.
- Sync basic data back to the iPhone app.
- Keep WatchOS app simple and reliable for MVP testing.

Phase 7: Testing, Optimization & Handover

The final phase will test the iPhone app, WatchOS companion app, BLE communication, haptic command flow, and session tracking.

Main focus:

- iPhone app testing.
- Apple Watch app testing.
- BLE connection testing.
- BLE reconnection testing.
- Live data streaming testing.

- Haptic command testing.
- Session tracking testing.
- Recovery rating testing.
- Battery/status display testing.
- Bug fixing.
- Final documentation and source code handover.

Milestones & Deliverables

Milestone	Deliverables
Milestone 1: App Scope & Architecture	App flow, MVP feature list, BLE communication plan, data flow, session logic
Milestone 2: UI/UX Design	iPhone and WatchOS screens for onboarding, pairing, dashboard, sessions, history, and settings
Milestone 3: iOS App Foundation	SwiftUI setup, navigation, reusable components, local storage, data models
Milestone 4: BLE Integration Layer	BLE discovery, pairing, connection handling, data receiving, haptic command sending
Milestone 5: Session & Insights Module	Session tracking, biometric dashboard, HR zones, recovery rating, session history
Milestone 6: Apple Watch Companion App	WatchOS session mirror, haptic reminders, HR zone display, sync with iPhone
Milestone 7: Testing & Handover	Tested iOS/WatchOS builds, bug fixes, BLE test notes, documentation, source code

Timeline

Phase	Estimated Hours	Timeline
Requirement Finalization & App Architecture	5 hours	Week 1
UI/UX Design	7 hours	Week 1
iOS SwiftUI App Foundation	9 hours	Week 1 – Week 2
BLE Communication Layer	14 hours	Week 2
Session Tracking & Recovery Rating	10 hours	Week 2 – Week 3
Lightweight Apple Watch Companion App	7 hours	Week 3
Testing, Optimization & Handover	8 hours	Week 3
Total Estimated Time	60 hours	3 Weeks

Total Timeline for App Development: 60 Hours

Work Breakdown Structure



KPI (Key Performance Indicator)

A measurable value that indicates how effectively the project is meeting its objectives.

Rate: 95%

Disclaimer: The KPI rate reflects the targeted performance level under planned project scope and resources. Actual outcomes may vary depending on unforeseen technical or operational challenges.

Creep (Scope Creep)

Uncontrolled or gradual expansion of project scope beyond its original objectives.

Rate: 5%

Disclaimer: The creep rate represents an estimated buffer for minor deviations. Any expansion beyond this rate will require formal change requests and client approval.

Timeline and Budget Justification

Timeline Justification

R and D

The 40-hour R&D timeline is structured to efficiently validate all critical technical aspects of the wearable prototype while minimizing development risk before full-scale implementation begins. The largest allocation is dedicated to biometric sensor and BLE validation because heart rate, SpO2, and especially GSR sensing require extensive experimental testing, filtering analysis, and motion-stability evaluation to achieve reliable wearable performance. Firmware and BLE integration also require significant development time due to real-time communication handling, fallback logic implementation, and power optimization testing. Smaller but essential time allocations are provided for haptic system evaluation, power subsystem validation, and wearability analysis to ensure the final prototype is technically feasible, comfortable for users, and stable enough to support future product development and investor demonstrations.

PCB

The 60-hour PCB development timeline is structured to ensure the creation of a compact, stable, and wearable-grade hardware platform while minimizing electrical, mechanical, and manufacturability risks. A significant portion of the timeline is allocated to biometric sensor integration and PCB routing because the project contains highly sensitive analog sensing circuitry, including GSR and optical biometric acquisition systems that require careful signal isolation, filtering, and grounding strategies. Additional time is dedicated to component placement optimization and power management design due to the strict wearable size constraints, battery safety requirements, and BLE antenna performance considerations. The routing and signal integrity phase receives the highest allocation because stable wearable operation depends heavily on EMI reduction, analog/digital separation, power stability, and optimized trace design. Final verification and manufacturing preparation hours ensure the PCB is fully validated, fabrication-ready, and suitable for reliable prototype assembly and future iteration.

Mechanical Design

The 60-hour mechanical timeline is planned to cover the complete prototype design cycle in a controlled and practical way. The first 15 hours are allocated to the mechanical concept and component layout to define the wristband form factor, internal spacing, sensor placement,

PCB/battery position, strap integration, and overall usability. The next 15 hours are assigned to enclosure CAD design, where the outer body, top cover, bottom housing, mounting points, USB-C access, and haptic motor area will be modeled. Another 15 hours are reserved for the bottom sensor-contact design to ensure proper skin contact, comfort, stable sensor readings, and ergonomic fit during wearing. The final 15 hours are kept for testing review and design refinement, including checking assembly fit, enclosure strength, sensor alignment, manufacturability, and making final CAD adjustments before prototype fabrication. This timeline is sufficient for a focused mechanical prototype stage while keeping the work efficient and aligned with the 3-week development plan.

App Development

The estimated mobile app development timeline is 3 weeks / 60 hours, based on a lean MVP-level iPhone and Apple Watch companion prototype. The scope is reduced to the essential features needed to validate the product concept: BLE pairing, live biometric display, session tracking, battery/connection status, haptic reminder command, basic session history, recovery rating, and a lightweight Apple Watch companion view.

The BLE communication layer receives the highest time allocation because it is the most critical part of the app. The app must reliably connect with the wristband, receive real-time biometric data, show device status, and send haptic commands back to the wearable.

The Apple Watch app is included only as a lightweight companion experience within this estimate. It will mirror the active iPhone session, show basic status, and support simple haptic reminders. A fully independent WatchOS app, advanced sync logic, deep analytics, and production-level polish are not included in this 60-hour phase.

The UI/UX, session tracking, and recovery rating modules are intentionally kept simple. Advanced AI-based insights, backend/cloud infrastructure, App Store deployment, full health algorithm development, subscription features, and production-ready analytics are excluded from this MVP timeline. The main goal is to create a working app prototype that can support hardware testing, BLE validation, and client demo requirements.

Budget Justification

The proposed project is justified by the comprehensive scope and technical depth of AI-Powered Smart Hydration & Wellness Wristband. This rate covers the complete project lifecycle, including advanced research and system architecture definition, Component selection and validation, PCB and Schematics, wireless integration, App development, system testing, and preparation of detailed technical documentation. Sufficient time is allocated for iterations, debugging, and validation to ensure electrical safety, signal integrity, wireless reliability, and long-term performance in real-world. The budget also includes continuous client communication and post-delivery technical support for clarifications and minor refinements. The fixed-rate model ensures budget predictability, with advance payment supporting early resource allocation and final payment upon successful delivery, resulting in a balanced and value-driven budgeting framework.



Portfolio



Ultra-Slim Smart Healthband

Muhammad Bilal Javid

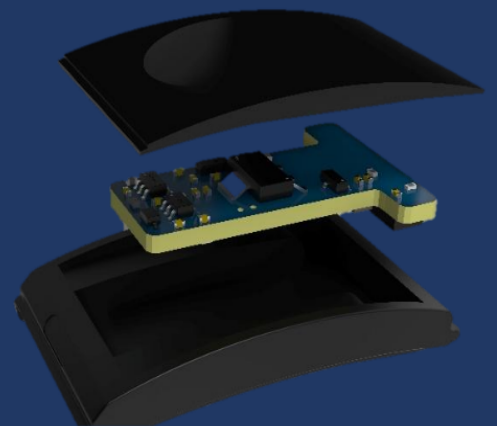


Problem

Modern health monitoring systems are often bulky, power-consuming, and lack real-time connectivity. Users need a wearable device that provides accurate biometric monitoring, long battery life, and seamless smartphone/cloud synchronization while remaining comfortable for all-day use. Existing wearables also lack reliable OTA updates and touchless control features.

Goal

- Design an ultra-slim smart health band for accurate biometric monitoring.
- Enable gesture-based touchless control and haptic alerts.
- Ensure seamless BLE 5.0 connectivity using Nordic nRF52840 SoC.
- Support OTA firmware updates for long-term reliability.
- Provide ultra-low power performance (2+ days per charge).
- Maintain comfortable 5.5 mm slim design for all-day wear.
- Include precision sensors (heart rate, SpO₂, etc.) and optional display.



The Ultra-Slim Smart Health Band is a sleek, reliable wearable designed for modern health monitoring. It combines Nordic nRF52840 BLE 5.0, precision biometric sensors, gesture control, and haptic feedback for accurate real-time tracking. With ultra-low power optimization and OTA update support, it provides long-lasting usability. The device is ideal for remote healthcare, fitness, and medical IoT applications.

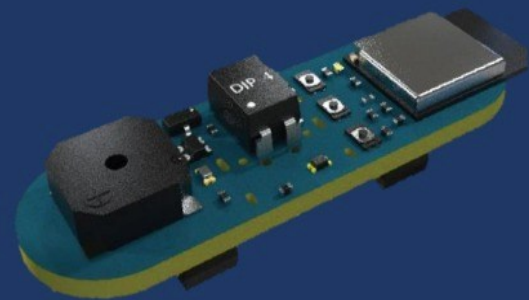
Smart Social WristBand

Muhammad Bilal Javid



Problem

Modern social interactions and networking events lack a discreet, low-latency method to broadcast social status, relying mainly on verbal communication and smartphone apps. Current solutions do not support secure proximity-based detection or privacy-preserving visibility. There is a need for a low-power wearable device that enables instant social cues with real-time synchronization and minimal user effort.



Goal

- Develop a BLE wearable status band for easy social interaction.
- Display social/relationship status using color-coded LEDs.
- Enable real-time status sync via mobile app using Nordic nRF52840 SoC (BLE 5.0).
- Provide privacy mode and discreet vibration alerts.
- Support onboard button control and optional GPS mapping/chat features.
- Ensure User-Friendly, Non-intrusive, and wearable Device



My Status Band is a BLE wearable designed to make social interactions more engaging and approachable. By integrating the Nordic nRF52840 SoC, color-coded LEDs, vibration alerts, and real-time app synchronization, it offers a fun and non-intrusive way to signal social availability. With privacy protection and optional GPS-based features, the band supports safe and meaningful connections in events, meetups, and social communities.

Pilgrimage Companion Device

Muhammad BilalJavid

Problem

Pilgrims performing Tawaf, Sa'i, and Tasbeeh often face challenges in tracking rituals accurately, accessing timely guidance, and managing orientation, especially in crowded or complex environments. Existing solutions may lack real-time responsiveness, GPS-based location accuracy, bilingual support, or reliable guidance, reducing convenience and precision.

Goal

- Develop AL-Mueen, an STM32-based, event-driven pilgrimage companion.
- Integrate a SH1106 OLED display for clear visual guidance and a DFPlayer audio engine for timely voice prompts.
- Utilize a GNSS/GPS parser for precise location and time updates.
- Offer a bilingual experience with instant switching between English and Arabic for text and audio.
- Ensure responsive interaction, smooth UI updates, and operational reliability in demanding environments.
- Provide accurate guidance for Tawaf, Sa'i, and Tasbeeh using a cooperative state machine and interrupt-driven inputs.

AL-Mueen delivers a responsive, bilingual, and GPS-enabled pilgrimage companion that guides users through Tawaf, Sa'i, and Tasbeeh with real-time visual and audio prompts. By combining STM32-based control, GPS positioning, an OLED display, and DFPlayer audio, the device ensures accurate, user-friendly, and consistent performance, enhancing the ritual experience for pilgrims.



Smart Insect Detection Camera System

Muhammad Bilal Javid



Problem

Accurate detection and observation of insects is challenging due to their small size, rapid movement, and susceptibility to image distortion in conventional camera systems. Existing solutions often lack sufficient optical precision, stability, or real-time AI processing capabilities, limiting their effectiveness in research, agriculture, and biodiversity.

Goal

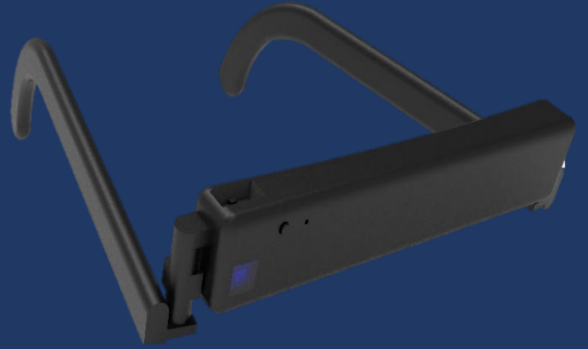
- Create a high-precision close-up imaging system for insects.
- Capture stable and distortion-free macro images.
- Integrate an embedded AI vision system for real-time detection.
- Use IMX219 8MP camera with a low-distortion lens setup
- Utilize MIPI-CSI interface for high-speed image transmission.
- Deploy RK3588 AI processor for on-device inference.
- Enable vibration-free imaging through optimized optical alignment.
- Provide AI/ML-ready real-time inference using TensorFlow Lite.
- Optimize system for field research and laboratory applications.



This project delivers a high-precision imaging and AI-powered insect detection system, combining optical accuracy with real-time machine learning. Using a low-distortion lens setup, IMX219 8MP camera, MIPI-CSI interface, and RK3588 processor, it enables stable close-up imaging and TensorFlow Lite inference. The system offers a reliable, scalable, and efficient solution for insect observation, research, and automated monitoring in both lab and field environments.

Smart Glasses

Muhammad Bilal Javid

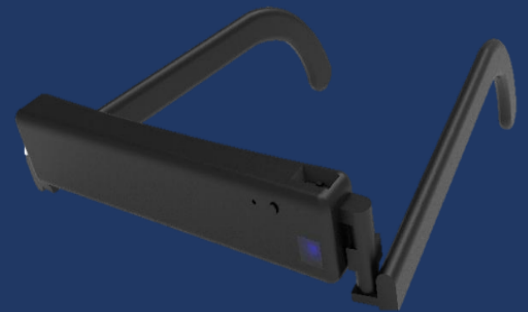
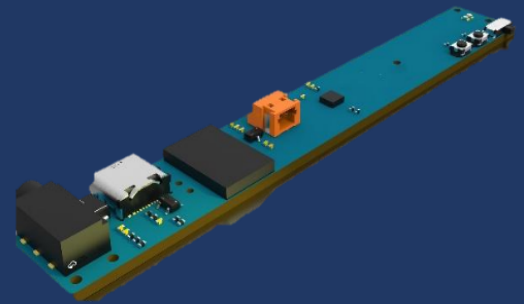


Problem

Individuals with physical disabilities often face challenges in interacting with computers or mobile devices due to limited mobility. Traditional input devices like mice and touchscreens are not always accessible, and existing assistive solutions may lack precision, responsiveness, wireless connectivity, or efficient powermanagement, limiting usability and convenience.

Goal

- Design and validate BLE-enabled smart glasses that control computer or mobile device cursors through head movements.
- Integrate a compact 4-layer PCB for a robust and space-efficient hardware platform.
- Utilize the MPU9250 gyro sensor for precise head movement tracking.
- Incorporate the BMD300 (nRF52832) Bluetooth module for reliable BLE connectivity.
- Implement a Battery Management System (BMS) for efficient charging, discharging, and thermal safety.



The BLE-enabled smart glasses provide an intuitive and accessible solution for handicapped individuals, enabling precise control of computers and mobile devices through head movements. By combining a compact 4-layer PCB, MPU9250 gyro sensor, BMD300 Bluetooth module, robust Battery Management System, and responsive interrupt-driven firmware, the device ensures seamless functionality, efficient power management, and user-friendly operation. This integration demonstrates a reliable and innovative approach to assistive technology.

Fitness Tracking System

Muhammad Bilal Javid

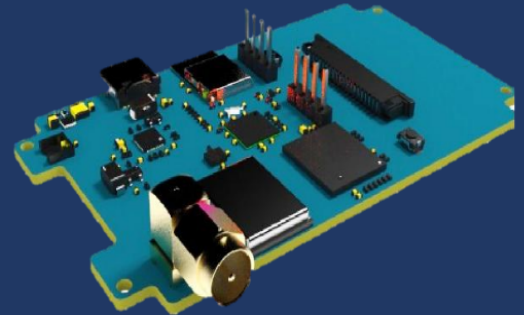
Problem

High-performance racing enthusiasts require accurate and real-time lap and section timing for performance analysis. Existing solutions often lack seamless wireless connectivity, efficient on-device processing, or reliable data transfer to companion apps, limiting usability, precision, and post-session analysis.

Goal

- Develop a Bluetooth-enabled racing display compatible with GPStracker such as the Crossbox CBX30 and RaceBox Mini-S.
- Enable real-time calculation and recording of lap times, section times, and session statistics directly on the device.
- Ensure seamless Bluetooth connectivity for data transfer to the Crossbox mobile app.
- Incorporate the BMD300 (nRF52832) Bluetooth module for reliable BLE connectivity.
- Design a compact, low-power PCB optimized for signal integrity, EMI control, and efficient wireless communication.
- Implement an STM32-based control architecture for stable, real-time processing and reliable device performance.

The Bluetooth-enabled racing display provides racing enthusiasts with a compact, reliable, and real-time timing solution. Its STM32-based architecture, optimized PCB design, and seamless Bluetooth connectivity allow accurate lap and section tracking, efficient on-device processing, and effortless integration with the Crossbox mobile app. This device demonstrates a robust and user-friendly approach to enhancing racing performance analysis.



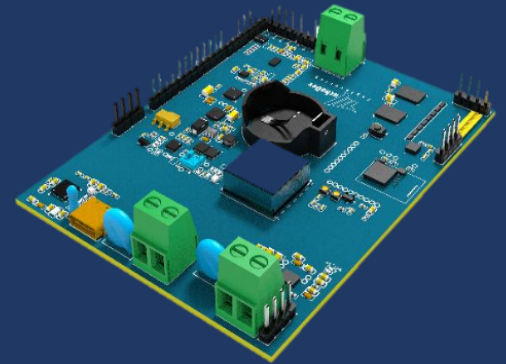
Smart Shoe

Muhammad Bilal Javid



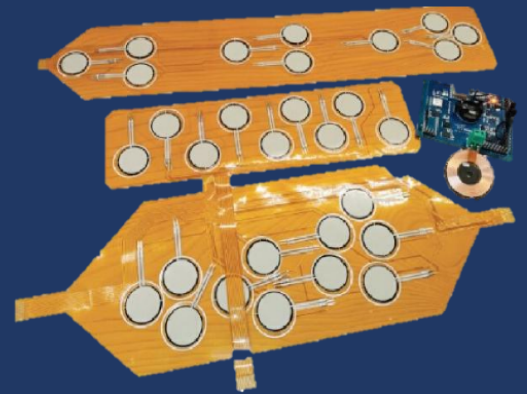
Problem

Patients with mobility impairments, stroke recovery, musculoskeletal disorders, or diabetes often lack real-time monitoring of foot pressure and gait patterns. This can lead to abnormal pressure distribution, increased risk of falls, delayed rehabilitation, and diabetic foot ulcers. Traditional monitoring methods are often expensive, non-continuous, and not suitable for long-term daily use.



Goal

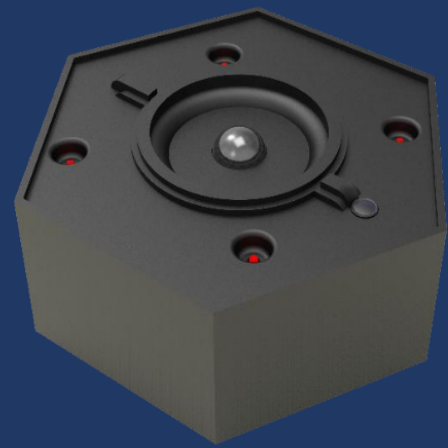
- Develop a smart shoe with flexible FSR sensor PCB for accurate foot pressure detection.
- Enable real-time monitoring of pressure distribution and gait patterns.
- Use Nordic nRF52832 (BMD-300) BLE module for low-power wireless data transmission.
- Support long-term wearability and comfort with a compact shoe-integrated design.
- Include wireless charging, motion/magnetic sensors, and mobile dashboard support.



This smart shoe provides a medical wearable solution for real-time foot pressure monitoring and gait analysis using a flexible FSR sensor PCB and Nordic nRF52832 (BMD-300) BLE module. With motion/magnetic sensors, battery power management, and wireless charging, it ensures accurate detection, low-power wireless connectivity, and comfortable long-term wear. The system supports fall detection and diabetic foot monitoring, enabling preventive healthcare and rehabilitation through continuous tracking and data insights.

Smart Fire Safety System

Muhammad BilalJavid



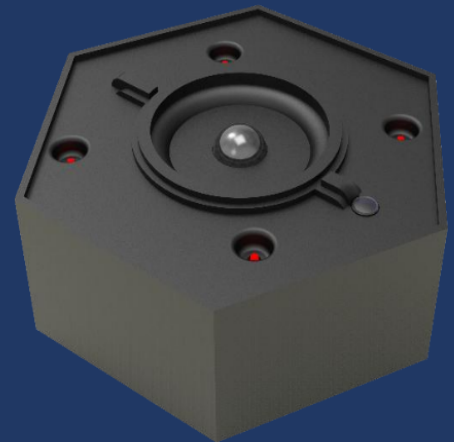
Problem

Fire hazards and poor indoor air quality pose serious risks to homes, offices, and industrial environments. Traditional fire safety systems often lack real-time monitoring, long-range communication, and intelligent alerts. Many solutions depend heavily on internet connectivity, have limited user interaction, and fail to provide early warnings for air quality degradation—leading to delayed responses and increased damage or safety risks.



Goal

- Provide a reliable, wireless, and intelligent safety solution.
- Instantly detect smoke, fire, and temperature changes.
- Continuously monitor indoor air quality for early hazard detection.
- Operate reliably even without internet connectivity.
- Ensure long-range and stable communication using LoRa mesh networking.
- Offer an intuitive touchscreen interface for monitoring and manual control.
- Comply with EMI/EMC standards for safe and robust operation.



The Smart Fire Safety and Monitoring System successfully addresses modern safety challenges by combining advanced sensors, long-range wireless communication, and an interactive user interface. With real-time alerts, offline functionality, and robust design, it provides a dependable solution for homes, offices, smart buildings, and industrial environments. This system enhances early detection, improves response time, and significantly increases overall safety and reliability.

Smart Luggage

Muhammad Bilal Javid



Problem

Traditionally, luggage lacks modern security, real-time tracking, and weight awareness, making travel stressful and unpredictable. Zipper-based locks are vulnerable to tampering, lost luggage is difficult to recover, and travelers often face unexpected overweight baggage fees due to the absence of built-in weight monitoring and smart alerts.



Goal

- Design a smart, connected luggage system to enhance travel safety, convenience, and peace of mind.
- Integrate a magnetic fingerprint lock for secure, tamper-proof access.
- Integrate Arduino-based control for multiple peristaltic pumps.
- Incorporate an embedded digital weighing system to monitor luggage weight in real time.
- Provide remote monitoring and control through the Smartie mobile application.
- Deliver real-time alerts for location updates, arrival notifications, and overweight warnings.



The smart luggage solution successfully modernizes travel by combining secure biometric locking, real-time global tracking, and intelligent weight monitoring in a single connected product. With mobile app control and instant notifications, the system minimizes travel risks, prevents overweight baggage issues, and provides travelers with confidence and control throughout their journey. This project demonstrates a practical, scalable approach to intelligent travel gear.

Smart Humidifier

Muhammad Bilal Javid

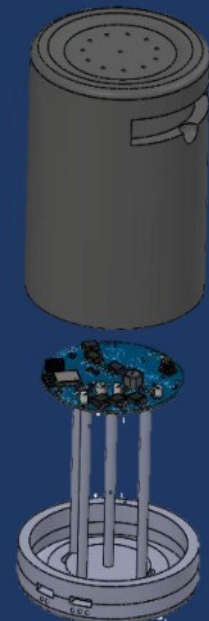


Problem

Traditional aromadiffusers and incense burners often lack precise temperature control and advanced safety mechanisms, leading to risks such as overheating, inconsistent burning, and potential fire hazards. Many devices also operate without real-time monitoring or smart control, limiting user awareness and integration with modern smart home systems. Additionally, inefficient hardware designs result in higher power consumption and reduced

Goal

- Develop anIoT-enabled home automation system using ESP32.
- Enable remote control of lights, fans, and appliances via Wi-Fi or Bluetooth.
- Provide real-time monitoring and intelligent scheduling.
- Integrate a TFT LCD touchscreen for intuitive local control.
- Support offline operation for uninterrupted functionality.
- Improve energy efficiency through motion-based and single-light control.
- Enhance home safety with air quality monitoring and smoke alerts.
- Enable multi-zone lighting and fan speed control.
- Optimize system for field research and laboratory applications.



The ESP32-based IoT aroma diffuser successfully combines safety, precision, and smart functionality in a compact design. Through accurate temperature regulation, real-time monitoring, and intelligent protection mechanisms, the system ensures reliable and hazard-free operation. Its scalable architecture, low power consumption, and user-friendly interface make it well-suited for home, hospitality, and smart environment applications, offering a modern and safe solution for controlled aroma diffusion.

App of Wearable Device

Muhammad Bilal Javid



App Features

- Developed a companion mobile app for a smart wristband prototype, built for investor demonstration and feature validation. Enabled seamless Bluetooth communication with the wristband.
- Displayed real-time sensor data and device status.
- Build a cross-platform mobile application for real-time monitoring and alert notifications. Design a wearable, ergonomic enclosure for user comfort and daily use.



App of Smart Home Controller

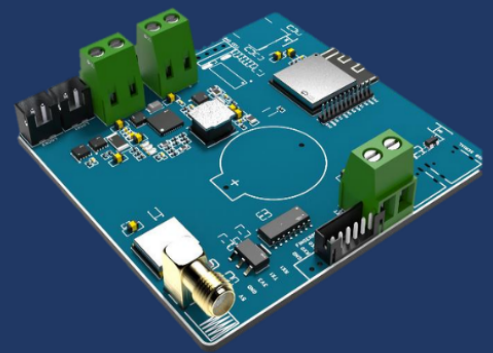
Muhammad Bilal Javid

App Features

- Developed a mobile app to interface with a next-gen humidifier control system powered by ESP32
- Enabled real-time monitoring and control of humidifier settings via Wi-Fi and Bluetooth
- Supported manual and scheduled relay control for connected components
- Integrated sensor data visualization for humidity, temperature, and system status
- Designed for BMS-friendly operation **with** Modbus and VFC-based triggers

App of Smart Luggage

Muhammad Bilal Javid



App Features

A companion mobile app for a smart luggage system focused on security and convenience.

- Enabled fingerprint-based and app-based unlocking via Bluetooth.
- Supported remote device management and instant notifications.
- Included user authentication, role-based access, and automated report generation.
- Designed for scalability, reliability, and compliance monitoring.

App of Smart Device Admin Panel

Muhammad Bilal Javid

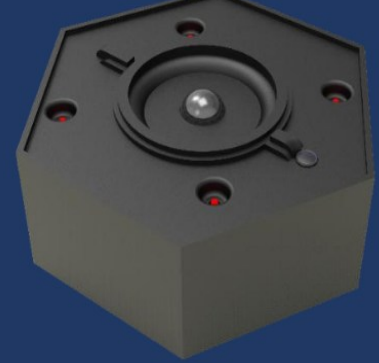
App Features

- Designed an admin-facing web app to manage and monitor smart home devices remotely.
- Enabled device registration, user management, and system diagnostics.
- Integrated with cloud backend for real-time status updates and troubleshooting.
- Provided usage analytics, firmware control, and alert notifications.
- Built with a responsive design for desktop and tablet use.
- Focused on reliability, control, and seamless backend



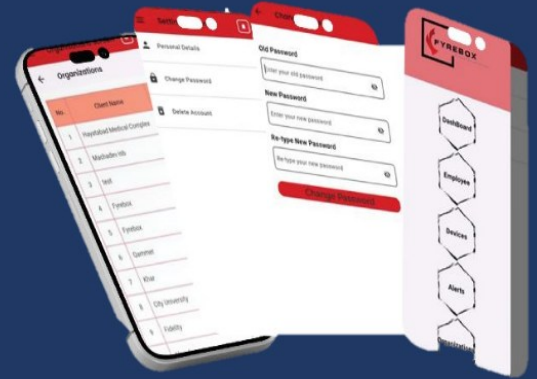
App of Fire Management

Muhammad Bilal Javid



App Features

- Developed a centralized web application for managing system Deployments monitoring.
- Implemented real-time dashboards for alerts, sensor data, and device status
- Supported remote device management and instant notifications.
- Included user authentication, role-based access, and automated report generation
- Designed for scalability, reliability, and compliance monitoring



App of Smart Humidifier

Muhammad Bilal Javid



App Features

- Developed a mobile app to interface with a next-gen humidifier control system powered by ESP32
- Enabled real-time monitoring and control of humidifier settings via Wi-Fi and Bluetooth
- Supported manual and scheduled relay control for connected components
- Integrated sensor data visualization for humidity, temperature, and system status
- Designed for BMS-friendly operation **with** Modbus and VFC-based triggers



Client Feedback

Technology Officer for Electronic Product Development

Great to work with Muhammad and his team



Project Budget: \$4,550

Pcb designer

Very professional team.



Project Budget: \$5,250

PCB Layout

Muhammad and his team executed the project flawlessly with clear communication, quick responses, and efficient progress tracking. They handled changes smoothly and consistently aimed for fast solutions. Highly recommended for future projects.



Project Budget: \$12,000

CAD design- Prototype

Muhammad and his team managed the project flawlessly from start to finish, with clear communication, fast responses, and excellent transparency via Google Docs. They efficiently handled changes and consistently delivered quick solutions. The client highly recommends them and would gladly work with them again.



Project Budget: \$3,500

PCB and Electrical design for a photoreactor

I enjoy working with this group. They are talented.



Project Budget: \$15,324

Electrical Engineer - IoT Kapton Heater Controller

Bilal and his team did good work. They are quick and communicate well, and are accommodating for any issues.



Project Budget: \$3,658

ESP32 Developer Needed for Feature Enhancement

Excellent team. They are a decent sized team with lot of PCB development experience and firmware development. I worked with a team before and this team has exceeded my expectations. Will use again when I need. Thanks for all the help and assistance.



Project Budget: \$8,759

Arduino/RC32 Developer for GPS Based Alert System for Airsoft Game

It was a pleasure working with this team. Great service and product!



Project Budget: \$5,823

Advance Existing Prototype - for Ultra Small Smart Flowmeter

Muhammad and his team delivered high-quality work by downsizing the PCB and choosing excellent components. Although firmware delivery took longer than expected, their accommodating nature and ability to bring in expert help led to a successful and satisfying project outcome.



Project Budget: \$18,000

PCB Layout

Muhammad and his team managed the project flawlessly from start to finish, with clear communication, quick responses, and transparent progress tracking. They handled changes efficiently and delivered fast solutions. Highly recommended for future collaborations.



Project Budget: \$5,000

Smart Alarm Clock With E-ink Display

Fast response time, good planning from the start and good support. It was a pleasure to work with the team, and I hope to rehire soon again.



Project Budget: \$3,000

Help Build Supplement Vending Machine Prototype

Muhammad Bilal and his team are very diligent, creative and hard working. their ability to be able to problem solve and create solutions for work moving forward has enabled the success of our project now and for the future. I would recommend to anyone.



Project Budget:\$ 5,000

Schematic Design for Arduino Based Project with Stepper Motors

Exceptional product and service



Project Budget: \$17,312

Automated Electric Shaver Development for Back Neck/Hairline

The team is extremely responsive and thorough. Highly recommend



Project Budget: \$3,658

Mechanical Engineer Needed for Steel Bullet Trap Design with Electronics Integration

Highly skilled and responsive, delivering quality work with great attention to detail.



Project Budget: \$14,310

Thank You!

200+

Clients.

100+

Project.

30+

Countries.

98%

Response rate

Thank you for your time and consideration. We would be happy to discuss our proposal further, address any questions you may have, and provide additional insights.

We look forward to your feedback and the opportunity to collaborate on the next steps.

CEO

MUHAMMAD BILAL JAVAID